

# Common Admission Test (CAT 2025)

## Official Detailed Syllabus & Blueprint: Mathematics (Test Code 101)

Curriculum: NCERT Class XI &amp; XII Standard

Level: Undergraduate &amp; 5-Year Integrated M.Sc

Session: 2025

| Parameter                             | Examination Specification                                                         |
|---------------------------------------|-----------------------------------------------------------------------------------|
| Examination Mode                      | Computer Based Test (CBT) Online Examination                                      |
| Total Questions (Mathematics Section) | 90 Multiple Choice Questions (Highest single weightage in CUSAT CAT)              |
| Marking Scheme                        | +3 marks for each correct response; -1 mark for each incorrect response           |
| Maximum Marks (Mathematics)           | 270 Marks                                                                         |
| Ranking Criteria                      | Mathematics score is the mandatory tie-breaker in CUSAT engineering/science ranks |

## Class XI Syllabus Modules

### Unit 1: Sets, Relations, and Functions

- **Set Theory:** Sets and their representations, empty set, finite and infinite sets, equal sets, subsets, power set, universal set. Venn diagrams, union and intersection of sets, difference of sets, complement of a set, properties of complement.
- **Relations & Functions:** Ordered pairs, Cartesian product of sets, number of elements in the Cartesian product. Definition of relation, domain, codomain and range of a relation. Functions as special relations, domain, codomain, range of real functions, constant, polynomial, rational, modulus, signum, exponential, logarithmic, and greatest integer functions.

### Unit 2: Complex Numbers & Quadratic Equations

- **Algebra of Complex Numbers:** Complex numbers as ordered pairs of reals, algebraic properties, Argand plane and polar representation. Modulus and conjugate of a complex number, triangle inequality. Fundamental theorem of algebra, solutions of quadratic equations with real coefficients in the complex number system, relation between roots and coefficients.

### Unit 3: Matrices and Determinants

- **Matrix Algebra:** Matrices, algebra of matrices, types of matrices, symmetric and skew-symmetric matrices, orthogonal matrices. Determinants of order up to  $3 \times 3$ , properties of determinants, minors, cofactors, adjoint and inverse of a matrix. Solution of systems of linear equations using Cramer's rule and matrix inverse method, consistency and inconsistency.

### Unit 4: Permutations, Combinations & Binomial Theorem

- **Combinatorics:** Fundamental principle of counting, factorial  $n$ , permutations and combinations, derivation of formulae for  $nPr$  and  $nCr$ , simple applications. Binomial theorem for positive integral index, general and middle terms, properties of binomial coefficients and applications.

### Unit 5: Sequences and Series

- **Progressions:** Arithmetic Progression (AP), arithmetic mean (AM), Geometric Progression (GP), general term of GP, sum of  $n$  terms of GP, infinite GP and its sum, geometric mean (GM), relation between AM and GM. Sum to  $n$  terms of special series: sum of  $n$  natural numbers, squares, and cubes.

### Unit 6: Mathematical Reasoning & Linear Inequalities

- **Logic:** Mathematically acceptable statements, connecting words/phrases "and", "or", "implies", "implied by", "if and only if". Validating statements using method of contradiction, contrapositive. Linear inequalities in one and two variables, graphical solution of systems of linear inequalities.

## Class XII Syllabus Modules

### Unit 7: Limit, Continuity, and Differentiability

- **Differential Calculus:** Real-valued functions, algebra of functions, polynomials, rational, trigonometric, logarithmic and exponential functions. Inverse trigonometric functions and their properties. Limits, continuity, and differentiability. Derivative of sum, difference, product, and quotient of functions. Derivative of trigonometric, inverse trigonometric, logarithmic, exponential, composite and implicit functions; derivatives of order up to two. Rolle's and Lagrange's Mean Value Theorems.
- **Applications of Derivatives:** Rate of change of quantities, monotonic increasing and decreasing functions, tangents and normals, maxima and minima of functions of one variable.

## Unit 8: Integral Calculus

- **Definite & Indefinite Integrals:** Fundamental integrals involving algebraic, trigonometric, exponential, and logarithmic functions. Integration by substitution, by parts, and by partial fractions. Integration using trigonometric identities. Fundamental theorem of calculus. Properties of definite integrals. Evaluation of definite integrals, determination of areas of regions bounded by simple curves in standard form.

## Unit 9: Differential Equations

- **Ordinary Differential Equations:** Ordinary differential equations, order and degree, formation of differential equations. Solution of differential equations by separation of variables method, solution of homogeneous and linear differential equations of first order ( $dy/dx + Py = Q$ ).

## Unit 10: Coordinate Geometry (2D and 3D)

- **Two-Dimensional Geometry:** Cartesian coordinate system in a plane, distance formula, section formula, locus. Slope of a line, parallel and perpendicular lines, intercepts of a line on coordinate axes. Various forms of equations of a line, intersection of lines, angles between two lines, distance of a point from a line. Standard forms of equations of a circle, radius, centre, equation of tangent. Conic sections: Parabola, ellipse, and hyperbola in standard forms, eccentricity, directrix, focus, and latus rectum.
- **Three-Dimensional Geometry:** Coordinates of a point in space, distance between two points, section formula, direction cosines and direction ratios. Angle between two lines, skew lines, shortest distance between two lines. Equation of a line in space.

## Unit 11: Vector Algebra

- **Vectors:** Vectors and scalars, addition of vectors, components of a vector in 2D and 3D space, scalar multiplication, dot product, cross product, scalar and vector triple products, geometrical applications.

## Unit 12: Statistics and Probability

- **Descriptive Stats:** Measures of dispersion: calculation of mean, median, mode of grouped and ungrouped data; calculation of standard deviation, variance, and mean deviation for grouped and ungrouped data.
- **Probability:** Probability of an event, addition and multiplication theorems of probability, Bayes' theorem, probability distribution of a random variable, Bernoulli trials, and binomial distribution.

## Unit 13: Trigonometry

- **Identities & Equations:** Trigonometric identities, functions, formulas for compound angles, multiple and sub-multiple angles. General solutions of trigonometric equations. Properties of triangles, inverse trigonometric functions and equations.